CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

May-2014

MA 203: Engineering Mathematics-IV

Date: 02.05.14, Friday **Time: 10:00 a.m. to 01:00 p.m.** **Total: 70 Marks**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 (4+3)

(a) Find Fourier transform of the function $e^{-kt^2}, \ k > 0$

(b) Find Fourier Sine and Cosine transform of the function $f(t) = 2e^{-5t} + 5e^{-2t}$

Q-2 (4+5+5)

(a) Prove that $\overline{z}$ is not differentiable.

(b) Show that the transformation $w = z + \dfrac{a^2 - b^2}{4z}$ is a circle whose centre is origin and radius is $\dfrac{a+b}{2}$ in z-plane into the ellipse of semi axes a and b in w-plane.

(c) Define Analytic function. Test the analyticity of function $\sinh z$ and if yes then find its derivative.

OR

Q-2 (4+5+5)

(a) Define Harmonic function. If $f(z) = u + iv$ is an analytic function then prove that real and imaginary parts of $f(z)$ are Harmonic.

(b) State and prove Cauchy-Riemann equations in polar form also find and Analytic function and its imaginary part if real part is given by $-r^3 \cos 3\theta$

(c) Find the image of circle $|z| = \lambda$ under the mapping $w = \sqrt{2}\,(1+i)\,z$ also sketch the image

Q-3 (4+5+5)

(a) Evaluate: $\displaystyle\int_{c|z|=1.7} \frac{\cos \pi z^2}{(z-1)(z-2)}\, dz$

(b) Evaluate line integral: $\displaystyle\int_C \bar{z}\, dz$ from $z=0$ to $z=4+2i$ along the curve C given by path $z=t^2+it$

(c) By method of Residues, obtain the inverse Laplace transform of $\dfrac{4s+5}{(s+2)(s-1)^2}$

OR

Q-3 (4+5+5)

(a) Evaluate: $\displaystyle\int_C \frac{z^3-z}{(z-2)^3}\, dz$ where C is ellipse $16x^2+25y^2=400$

(b) State Cauchy residue theorem and hence evaluate: $\displaystyle\int_{c|z|=2} \frac{2z-1}{z(z+1)(z-3)}\, dz$

(c) Obtain Laurent's series expansion of function $f(z)=\dfrac{1}{4z-z^2}$ in the regions $0<|z|<4$

SECTION-II

Q-4 (3+4)

(a) Define unit impulse function and find its Z-transform.

(b) Find the Z-transform of $\sin(3k+5)$

Q-5 (7+7)

(a) Using the Z-transform, solve $y_{n+2}+4y_{n+1}+3y_n=3^n$ with $y_0=0, y_1=1$.

(b) Calculate the approximate value of $f(x)$ for $x=46$ & $x=63$ using the table:

x	45	50	55	60	65
$f(x)$	114.84	96.16	83.32	74.48	68.48

OR

Q-5 (7+7)

(a) From the following table find $f(x)$ and hence $f(6)$ using Newton's divided difference formula:

x	1	2	7	8
$f(x)$	1	5	5	4

(b) Solve $x-\ln x=0$ correct up to three decimal places by false position method.

Q-6 (7+7)

(a) Applying Bisection method, find the smallest positive root of $f(x)=xe^{-x}$ by performing five iterations.

(b) Using appropriate Simpson's rule, evaluate $\int_{-\pi}^{\pi} e^{\sin x}\,dx$, by taking $h=\frac{\pi}{4}$

OR

Q-6 (7+7)

(a) Use the Runge-Kutta fourth order method to find $y(0.2)$ with $h=0.1$ for the initial value problem $\frac{dy}{dx}=\sqrt{x+y}$, $y(0)=1$.

(b) Using Newton-Raphson method to find the smallest positive root of the equation $\sin x = x\cos x$ correct up to four decimal places.